ClaimLedger: from idea to local MVP

Product, implementation and pilot plan | Updated 5 October 2026

Product promise
Turn a grocery shortage deduction into a checked evidence packet, show what is missing, and track the actual retailer credit through resolution.

The problem remains specific

A supplier invoices a retailer, but receives less money after an alleged delivery shortage. The finance team must connect the deduction to a shipment, locate the invoice and transport records, check quantities and prepare a challenge. The evidence may be scattered across spreadsheets, warehouse scans, carrier files and retailer portals.

The UK Groceries Code Adjudicator's 2026 survey received more than 2,600 responses: 20% reported problems with processes for resolving invoice discrepancies, 14% reported payment delays, and 32% experienced a Code issue. These are reported supplier problems, not invoice error rates or proof that shortage deductions are invalid. [1]

What exists today

A working local, single-operator web app now supports imports, original-document storage, per-SKU evidence review, local OCR, draft/reviewed PDF and ZIP packets, submission records, decisions and actual credits. React, TypeScript and Tailwind power the interface; Python and SQLite hold the workflow and persistent records.

Built and checked	Still a hypothesis
59 backend tests plus 6 frontend request tests; strict type check, production build and browser checks.	Real supplier time savings, retailer acceptance, recovery uplift and willingness to pay.
Bulk uploads, shared multi-SKU originals, independent reviews and OCR source highlights.	A repeatable advantage over established deduction tools and a scalable 100-customer service.

The next decision

Use the MVP to learn from 3-5 established suppliers and 30-50 redacted historical shortage cases. Select one retailer and country from actual customer access. The current release is a pilot workbench, not a retailer-certified or hosted SaaS product.

Pricing hypothesis: USD 180-250 per business per month; test USD 229 as the initial anchor. Pricing, processing allowances and onboarding effort need customer evidence. At 100 customers, USD 229 would equal USD 22,900 monthly recurring revenue before costs. This is a target, not a forecast.

Reading guide: workflow and features (2-4); actual architecture (5); validation (6); market and economics (7); pilot and roadmap (8); operation and sources (9).

The customer and the daily workflow

One finance team, one deduction line, one reviewable evidence trail

Who buys, who uses, and when

Initial customer hypothesis: an established small or medium grocery brand or supplier with regular retailer transactions and recurring shortage deductions. An owner, finance manager or controller can approve the purchase; an accounts receivable administrator assembles cases. Avoid businesses with too few monthly cases to justify a subscription.

Prioritize English-speaking teams in the UK, Canada and US, with Ireland or other English-speaking European teams as later possibilities. The current app has no country-specific policy engine or validated retailer integration. GBP, CAD, USD and EUR support is currency handling, not proof of market or regulatory coverage.

Step	Operator action	Useful output
1. Capture	Upload a CSV/XLSX export or enter a deduction manually.	A persistent case with references, amount, owner and original import row.
2. Investigate	Upload originals; inspect reference/SKU matches and source pages.	Suggested matches plus a clear list of missing or conflicting evidence.
3. Review	Confirm invoice, PO, SKU, quantity, unit, page and receipt qualification.	An independent reviewed mapping for this case line.
4. Prepare	Generate a draft, or review and generate a final packet.	A PDF summary and ZIP of unchanged originals with a manifest.
5. Resolve	Submit outside the app; record the reference, decision and actual credit.	Separate disputed, approved and credited amounts with correction history.

Worked example: do not overclaim the evidence

Illustrative only: a GBP 600 deduction alleges 20 missing cases. The invoice and dispatch record each show 100 cases, but the receipt acknowledges pallets only. ClaimLedger flags the qualification and cannot unlock a final packet. A signature alone is not proof of the SKU quantity.

If a reviewed receipt explicitly confirms 100 cases in the same unit and references reconcile, the consistency checks can become ready. The operator still checks the retailer's applicable submission requirements and deadline. Readiness does not establish authenticity, liability or acceptance.

If the retailer later approves GBP 400 and a remittance actually credits GBP 250, the app records GBP 400 approved, GBP 250 credited and GBP 150 awaiting credit. It does not call the approval recovered cash. These figures illustrate the workflow; they are not customer results.

What changed from the original proposed brief

Shared multi-SKU originals are implemented, but each case still represents one SKU and one shipment. Split deliveries, a public upload link, an external extraction service, a policy-version engine and a durable background job queue were proposed earlier and are not in the working release.

Implemented MVP: intake and investigation

Practical automation, with human confirmation at the evidence boundary

Capability	What is implemented	Why it matters
Deduction intake	CSV/XLSX column mapping, preview, full-batch validation and duplicate detection. Manual case creation/editing; original row provenance retained.	Catch malformed data before saving and avoid overwriting existing cases.
Case organization	Owner, notes, explicit deadline source, follow-up date, search, currency selector, readiness filters and activity trail.	A finance manager can identify the next action without opening every file.
Original library	PDF, PNG, JPEG and TXT uploads; workspace-specific SHA-256 deduplication. Originals remain unchanged.	One file can be reused without creating a separate copy for every SKU.
Bulk upload	Up to 50 files per batch; each saves independently. Per-file failures and duplicates appear in the results.	Successful uploads survive another file failing.
Exact-match suggestions	Explicit invoice/PO/SKU references produce reviewable suggestions with a reason. Conflicting SKUs are excluded from exact-match suggestions.	Matching assists navigation; it does not confirm a shipment automatically.
Independent line reviews	Quantity, unit, source page, reviewer, notes and review version belong to the case/document link.	Reviewing or unlinking one SKU does not change another SKU on the same original.
Text and local OCR	PDF/TXT extraction and RapidOCR/ONNX for image/scanned pages. Source text, model scores and image boxes appear beside the original.	Inspect the source location and correct a suggestion before it becomes evidence.

OCR is assistance, not an approval system

Recognition runs locally; documents are not sent to an external AI API. Low-score spans are marked, but high scores still require review. Bulk OCR reads the first page of scans; other relevant pages are processed on demand. Candidate parsing supports labelled fields and delimited SKU/quantity/unit rows. Arbitrary retailer tables still need manual mapping.

Changing the selected case line, source page or candidate clears the verification checkbox. Confirmed source snippets are copied into the reviewed mapping, so rerunning OCR cannot silently rewrite the evidence that supported a prior review.

Current processing limits

Imports: 2,000 rows and 5 MB. Evidence: 10 MB per file, 100 MB per bulk batch, up to 100 PDF pages. Expanded XLSX content is limited to 30 MB. Limits are safety/processing constraints, not a commercial plan allowance. No large-volume throughput or recovery SLA has been established.

Implemented MVP: evidence to actual credit

The backend controls readiness, versions, packet snapshots and money

What the evidence checks establish

- The case has invoice, purchase order and SKU references. Financial evidence must agree on those references and the quantity unit; no implicit conversions occur.

- An invoice, dispatch record and delivery receipt are reviewed with numeric quantities and valid source pages. The full-delivery argument requires quantities to reconcile and a positive shipment quantity.

- The receipt explicitly substantiates the SKU count. Pallet-only, subject-to-count, shortages noted, missing or contradictory evidence require investigation.

- The alleged shortage is positive and does not exceed the invoiced quantity. Unknown deadlines remain unconfirmed; entered deadlines require a recorded source.

State or action	Rule in the current release
Open / ready for review	Readiness is an assessment of an open case, not a separate financial outcome. A draft may be exported with labelled gaps.
Reviewed final packet	Passing evidence checks plus a named reviewer and explicit verification are required. PDF/ZIP exports are saved snapshots.
Submitted	The operator records a retailer submission already made. A current final packet is required; a stale export cannot unlock submission.
Approved / rejected	Record the retailer response and source. Partial approvals are supported. An approved case can still have zero actual credits.
Accepted	An open case may be closed as accepted with a reason when the deduction is valid or the operator chooses not to dispute it.
Actual credit / credited	Record a verified remittance or credit-note allocation. Credits cannot exceed remaining approval; full credit closes the approved balance.
Correction	Reverse an erroneous credit entry with a reason. The original remains visible and the approved balance reopens. Closed decisions do not yet have a reopening workflow.

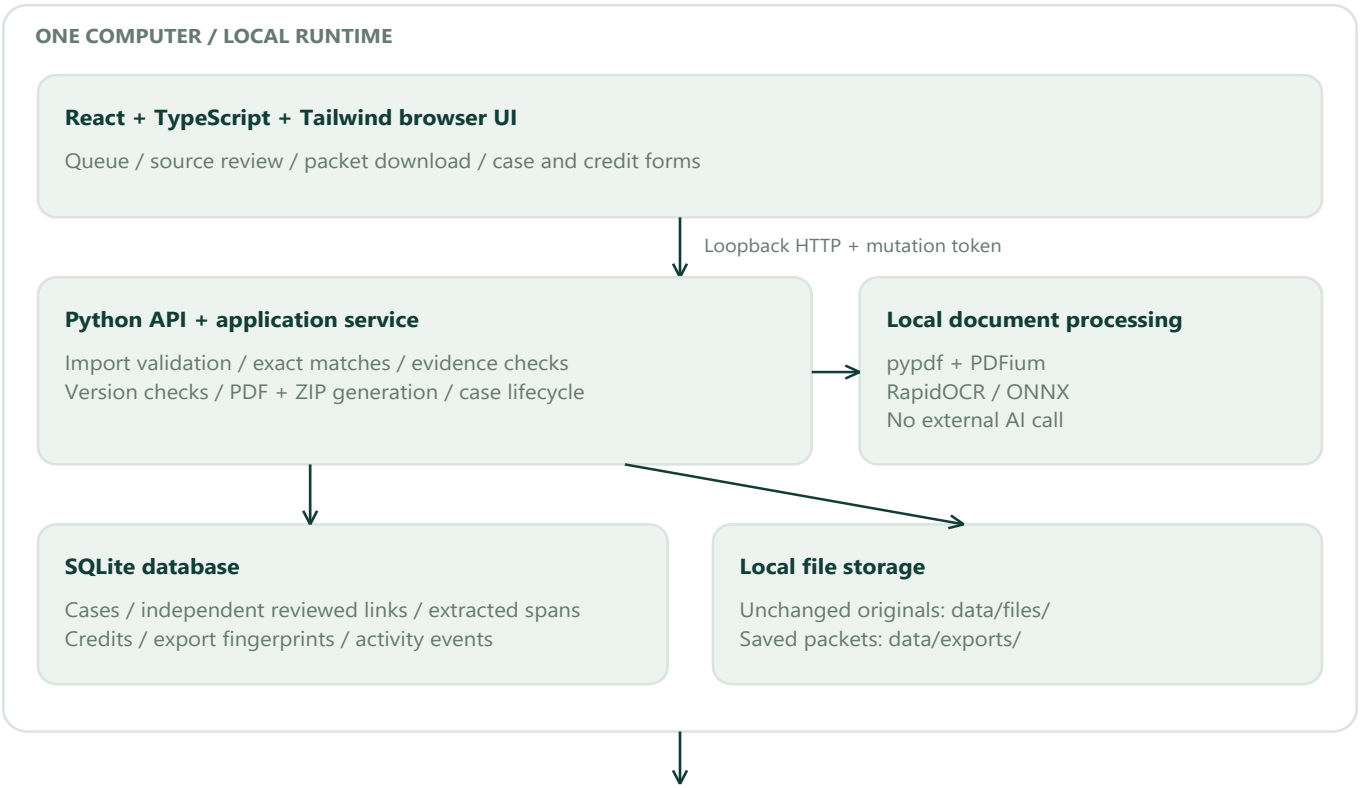
Integrity rules that matter operationally

Amounts use integer minor units. Totals are shown separately by currency and are never converted or combined implicitly. Import commits are atomic; stale case and mapping versions are rejected. Confirmed original hashes and packet fingerprints help detect changed evidence. These are consistency controls, not tamper-proof certification against someone with filesystem access.

Credit references are unique within a workspace. For a credit covering several deductions, use distinct allocation references and retain the original source reference. Each case has an approval cap, but the app cannot verify the original credit's total across those allocations; the operator must reconcile that total separately.

Actual architecture and storage model

The implementation replaces the original proposed cloud/worker design



Vite builds the included frontend assets; it is not a required production runtime. The normal launcher runs one Python process on 127.0.0.1. OCR is requested per page or after upload; there is no durable background worker, hosted OCR service or retailer-portal connection.

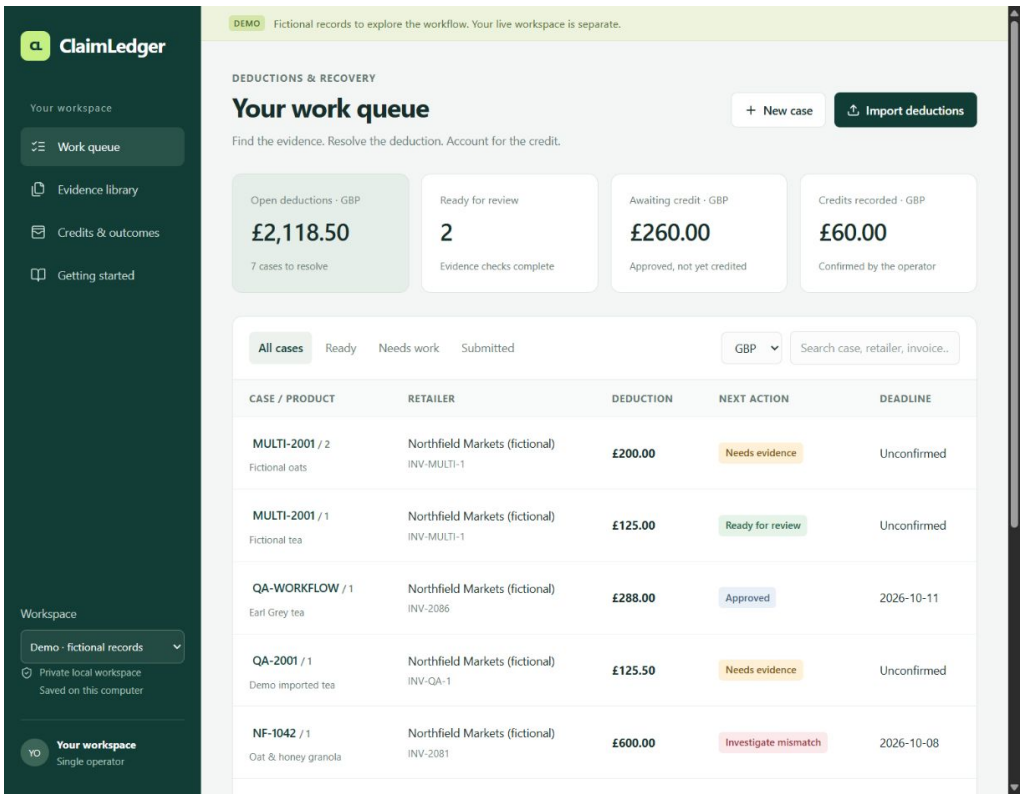
Data relationship	Responsibility
Case - reviewed link - original	A case is one deduction line. An original can support several cases; each link retains independent reviewed fields, reviewer, version and frozen source snippets.
Original - extraction pages	The original hash/file is unchanged. Extraction pages store native/OCR spans, confidence and image boxes as reviewable suggestions.
Case - exports / credits / events	Exports retain a fingerprint and reviewer. Non-reversed credits determine credited totals. Events record workflow changes and correction reasons.

Local request protection is not hosted security

Host and Origin checks, a per-start mutation token, restrictive Content Security Policy, file-path containment and workspace checks protect local requests. Demo and Live are logical local workspaces, not SaaS tenants. There are no accounts, roles, encryption at rest, malware scanning, managed backups or public-deployment safeguards. Retailer credentials are not required.

What has actually been validated

Implementation evidence, based on fictional records and disposable test stores



Current React work queue. All displayed retailers, products and transactions are fictional demonstration records; the screenshot is not evidence of customer recovery.

Verification	Result and boundary
59 backend tests	Precision, imports, atomicity, evidence checks, line isolation, stale versions/packets, OCR, partial/concurrent credits, persistence and HTTP protections.
6 frontend request tests	Workspace/token capture, preview query isolation, conflict errors, abort handling, oversize evidence and currency formatting. Strict TypeScript and Vite build pass.
Browser and PDF checks	Real picker imports/uploads, original previews, SKU review, OCR highlights, packet export, over-credit errors, reversals and phone layouts. Generated packet pages were rendered and inspected.

SQLite integrity and foreign-key checks passed. Existing originals and data were verified after moving the project to Desktop. Development-mode proxy writes were checked. Test counts describe the last completed implementation checks; they are not a production uptime or security certification.

Not yet proven

No real supplier cases, paid customers, accepted retailer packets, measured preparation-time savings or attributable recovery uplift have been established.

Competitive position and subscription economics

A useful wedge must outperform an existing workflow, not just a blank spreadsheet

Alternative	Verified advertised capability	Implication for ClaimLedger
SupplyPike [2]	Document Explorer describes automated shipping-document retrieval/organization and OCR; the cited announcement is from 2022.	Document collection is an established problem. Manual uploads are a disadvantage when customers need retrieval automation.
Revya [3]	Advertises dispute documentation, follow-up and system integrations. Pricing starts at USD 500+ per month plus a recovery percentage.	A narrower workbench could be cheaper, but we have not established comparable retailer coverage or better results.
Shelf-to-Cash [4]	Advertises remittance categorization and dispute tracking; Pro is listed at USD 199/month.	The target price band already has competition. A dashboard or generic categorization alone is weak differentiation.

Vendor descriptions and prices were checked on 5 October 2026. They are marketing statements, not independently validated performance; scopes and commercial terms may change. Revya's FAQ is inconsistent about broad retailer coverage versus distributor-first focus, so coverage must be confirmed directly.

The advantage to test

Position the product as a quick-start evidence-preparation workbench for one chosen retailer: shared originals, independent SKU reviews, explicit quantity/qualification checks, referenced packets and actual-credit reconciliation. Local processing is a design choice; it becomes a selling point only if buyers value it. We have not demonstrated that competitors lack these features.

Choose the first UK, Canadian or US retailer from pilot access and common case patterns. The generic current app is not yet the proposed retailer-specific advantage. A validated evidence checklist, import preset and output format may be the first defensible layer.

Price should follow measurable work removed

At USD 229/month, saving 6 hours at an illustrative USD 40/hour equals USD 240 of staff time. Count incremental verified credits separately and avoid attributing credits the customer would have recovered anyway. Staff cost and case volume are customer-specific assumptions, not proven savings.

100-customer goal	Revenue before costs
USD 180-250 monthly price range	USD 18,000-25,000 monthly recurring revenue.
USD 229 anchor	USD 22,900/month; USD 274,800 annualized, assuming all 100 remain paying.

Support determines whether a solo business works: 20 minutes per account per month becomes about 33 hours at 100 customers; 60 minutes becomes 100 hours, before sales and development. Measure onboarding, evidence chasing, OCR corrections and portal work. These are planning calculations, not a profitability forecast.

Customer discovery, pilot gates and roadmap

Use the existing MVP to decide what deserves the next build

Stage	Proposed activity	Evidence required to continue
1. Access	Build a list of 40-60 qualified suppliers; aim for 10-15 finance interviews. Request recent shortage examples, not general feature wishes.	Recurring case volume, accessible source documents, a buyer with budget, and a costly preparation bottleneck.
2. Historical test	With consent, examine 30-50 redacted cases from 3-5 suppliers. Select one country/retailer. Compare existing preparation with MVP-assisted review.	Coverage of real layouts and exceptions; reliable reviewed quantities; packets usable with minimal rewriting.
3. Supervised pilot	Run across successive payment cycles. Confirm retailer requirements and deadline sources. Test USD 229/month or another price within the target band.	Recurring use, paid intent, recorded support time and actual retailer responses. At least three paying businesses is a proposed initial gate.

Measure preparation, not just extraction speed

Time the same task from available records to a reviewable packet, including lookup, OCR correction and human verification. Record median active preparation time, missing-document rate, ambiguous-match rate, reviewer correction rate, rework and monthly case volume. Track waiting for retailer responses separately. The MVP does not yet collect these pilot metrics automatically.

Proposed gates: at least 50% less active preparation time on supported cases; no silent acceptance of ambiguous evidence; packets usable without substantial rewriting; three paying customers with recurring use; and low ongoing support effort. Keep approvals, actual credited value and incremental attributable recovery distinct. These are targets to test, not achieved results.

Build next in the order customer evidence supports

- Before charging: one retailer import preset, a reviewed evidence/deadline checklist and retailer-compatible packet formatting. Verify these with the customer's current instructions; a country code alone is insufficient.

- When repeated friction appears: clearer exception handling, better table parsing, document-request templates and faster keyboard review. Add split shipments or unit conversions only after real cases justify a reconciled data model.

- Before hosted or team use: authentication, roles, tenant/file isolation, encryption, malware controls, managed backup/restore, queueing, observability and deployment hardening. This is a separate delivery scope.

Pause or change the wedge if documents are routinely unavailable, delay is mostly retailer response time, low claim volume prevents monthly value, integrations dominate support, or an existing product already solves the same workflow economically.

Operation, handover and source notes

Working local software, with explicit boundaries and a reproducible starting point

Where the MVP lives

Current project: C:\Users\Aaditya Bisaria\Desktop\ClaimLedger
Start: double-click Start-ClaimLedger.cmd and keep its terminal open.
Open: http://127.0.0.1:8765/. Avoid launching a second server on the same port.

Use Demo for exploration and Live for your own records. The ordinary launcher uses the included frontend build and the project Python environment. On a new machine, install Python and requirements.txt first. Frontend changes require Node 22.18+ and pnpm 11; build instructions are in frontend/README.md. Refresh the browser after rebuilding or restarting the server.

Data handling and recovery

Cases and reviews live in data/claimledger.sqlite3; originals in data/files/; packet snapshots in data/exports/. Stop the app before backing up the entire data folder and restore the full folder with the server stopped. There is no cloud backup, retention automation or encryption at rest. Access to the Windows account exposes the files. Do not expose this loopback development server publicly.

Scope boundaries to carry into a pilot agreement

Shortage deductions, one SKU/line and one shipment per case, with a full-delivery consistency argument. Shared multi-SKU originals are supported. Pricing, promotions, damage claims, split shipments, implied conversions, broad contract interpretation, legal escalation and ERP/retailer integrations remain outside automatic processing. Missing-evidence messages are copyable drafts only; there is no public upload link, automated email or portal submission.

Sources and verification basis

[1] UK Groceries Code Adjudicator - 2026 survey release, published 16 April 2026. Supplier-reported invoice-discrepancy and payment issues. Open source.

[2] SupplyPike / SPS Commerce - Document Explorer announcement, last updated 17 March 2022. Historical evidence of retrieval/OCR competition, not a current full feature audit. Open source.

[3] Revya - FAQ: advertised workflow, pricing and integration scope. Coverage should be confirmed directly. Open source.

[4] Shelf-to-Cash - Product/pricing page: categorization, tracking and USD 199/month Pro listing. Open source.

External sources checked 5 October 2026. The market evidence motivates discovery; it does not establish universal legal entitlements or any retailer deadline. No policy deadline in this brief is presented as legal advice.

Implementation sources: project README.md, VALIDATION.md, frontend/README.md, server.py and app/store.py; the original 3-page concept brief dated 4 October 2026. Validation records describe synthetic tests and demonstrations, not live customer results.

Key source files: frontend/src/ for React UI and typed API; app/imports.py for intake; app/evidence.py and app/extraction.py for review; app/ocr.py and app/preview.py for local source reading; app/service.py for lifecycle transactions; app/packets.py for PDF/ZIP; tests/ and frontend/tests/ for checks.